MODULE 29 · DOMAIN 3 - COST-OPTIMIZED ARCHITECTURE DESIGN

Cost Allocation, Tagging and Replication Cost

Design tags that finance can use, enforce them before resources exist, and replicate data across Regions only as much as recovery needs.

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Allocation starts with **accounts**, and tags fill in the rest. A tag nobody enforces is a tag nobody can trust.

A practical tagging strategy

TAG	ANSWERS	EXAMPLE VALUES
CostCentre	Who pays?	CC-4100, CC-4200
Environment	Which stage?	prod, staging, dev
Application	Which workload?	pricing, tracking
Owner	Who to ask?	team or role, never a person's name

ON THE EXAM

Few mandatory keys, fixed allowed values, defined once centrally.

From tags to the bill

- **Activate** user-defined tags as cost allocation tags in the management account.
- **AWS-generated tags**, such as createdBy, can be activated too.
- **Backfill** activation for up to twelve months of history.
- **Only where the tag existed**: backfill cannot invent past tags.
- **Then** Cost Explorer, CUR 2.0 and Budgets can group by the tag.

> ON THE EXAM
>
> Tags present but missing from Cost Explorer: they were never activated as cost allocation tags.

Enforcing tags: three tools, three jobs

TOOL	DOES	DOES NOT
`Tag policy`	Standardises keys and allowed values	Require a tag on most creates
`Tag policy, required tags`	Validates IaC deployments for required keys	Cover console or CLI creates
`SCP with aws:RequestTag`	Denies creation without a tag	Check values easily or fix old resources
`Config required-tags rule`	Finds untagged resources; can remediate	Prevent creation

ON THE EXAM

Consistent values: tag policy. Must exist at creation: SCP. Find and fix existing gaps: Config.

Accounts, OUs and cost categories

- **Account per team or product**: allocation without tags.
- **Cost categories** group accounts, tags and services into business units.
- **Rules apply to history**, so reorganisations do not need re-tagging.
- **Shared accounts** still need tags or split rules for their costs.

> ON THE EXAM
>
> Map accounts and tags to business units with cost categories, not by renaming tags.

What replication costs

REPLICATION	YOU PAY FOR
S3 Cross-Region Replication	Second copy, inter-Region transfer, requests
S3 Replication Time Control	Extra per GB for the 15-minute SLA
DynamoDB global tables	Replicated writes in every replica Region, storage
Aurora Global Database	Replicated write I/O, secondary clusters, transfer

ON THE EXAM

Every extra Region multiplies storage, writes and transfer. Add Regions only for a reason.

Replicating cost-consciously

- **Filter S3 replication** by prefix or tag: copy what recovery needs.
- **Cheaper destination class**, such as Standard-IA or Glacier tiers.
- **RTC only** when a 15-minute replication objective is required.
- **Fewest replica Regions** that meet recovery and latency needs.
- **Backups instead of live replicas** where recovery time allows.

> ON THE EXAM
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> Match replication to the recovery objective, not to every object or every Region.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Tags on resources, missing in Cost Explorer	Activate, then backfill
prod, Production and PRD all in use	Tag policy with allowed values
No resource without a cost centre	SCP on aws:RequestTag
Replicating everything, only some needed	Filtered replication, cheaper class
A replica Region nobody needs	Remove it
Business units span many accounts	Cost categories

ON THE EXAM

Separate the jobs: activate, standardise, require, detect, and group.

> Meridian Freight's resources have carried a CostCentre tag for the past eight months, but the tag does not appear in Cost Explorer.
>
> Finance wants costs grouped by CostCentre, including the last six months.

What should the solutions architect do?

A Activate CostCentre as a cost allocation tag in the management account, and request a backfill for the past six months

B Activate the tag in each member account

C Re-tag every resource with a new key

D Create a cost category without activating the tag

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Activate and backfill

Makes the tag visible, and applies it to past months where it existed.

B Activate per member account

Activation is done in the management account.

C Re-tag

The tags already exist; they are just not activated.

D Cost category only

Tag rules need the tag to be in the billing data.

THE SIGNAL

Tagged but invisible: activate. Need history: backfill, up to twelve months.

> Meridian Freight's cost reports show Environment values of prod, Production, PRD and live for the same thing. Teams tag through the console, the CLI and infrastructure as code.
>
> The company wants one approved set of values across the organisation, and non-compliant values blocked on supported resources.

What should the solutions architect implement?

A. A wiki page listing the approved values

B. A monthly script that renames tag values

C. A tag policy defining the Environment key's allowed values, with enforcement for the relevant resource types

D. An AWS Budgets alert per value

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Wiki page

A convention; the reports show it is not followed.

B Rename script

Cleans up after the fact, every month, forever.

C Tag policy

One definition of allowed values, enforced organisation-wide.

D Budgets alerts

Unrelated to tag consistency.

THE SIGNAL

Consistent values across the organisation: a tag policy.

> Meridian Freight requires every new EC2 instance and RDS database to carry a CostCentre tag when it is created, whether created by console, CLI or code.
>
> Creation without the tag must be blocked in every member account.

What should the solutions architect implement?

A. A Config required-tags rule with automatic remediation

B. A tag policy listing CostCentre

C. A cost category that groups untagged resources

D. An SCP denying ec2:RunInstances and rds:CreateDBInstance when aws:RequestTag/CostCentre is null

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A **Config rule**

Detects after creation; does not block.

B **Tag policy**

Standardises values; does not block untagged creation from every tool.

C **Cost category**

Reports untagged cost; blocks nothing.

D **SCP on RequestTag**

Denies the create call itself when the tag is missing.

THE SIGNAL

Block creation without a tag, from any tool: an SCP with a Null condition on aws:RequestTag.

> Meridian Freight replicates an entire 900 TB S3 bucket from eu-west-1 to eu-central-1 for disaster recovery, using Standard storage and Replication Time Control.
>
> Only the shipments/ prefix, about 150 TB, is needed for recovery, and a recovery point of a few hours is acceptable.

What should the solutions architect change?

A Turn off replication and rely on versioning in one Region

B Filter the replication rule to the shipments/ prefix, store replicas in a cheaper storage class, and remove Replication Time Control

C Replicate to a third Region as well

D Switch to Same-Region Replication

◆ PAUSE AND DECIDE

What should the solutions architect change?

A No replication

Loses Regional disaster recovery entirely.

B Filter, cheaper class, drop RTC

Copies one-sixth the data, stores it cheaply, drops an unneeded SLA.

C Third Region

Adds cost with no stated need.

D Same-Region

Does not survive a Regional outage.

THE SIGNAL

Replicate what recovery needs, to the cheapest class that recovery tolerates.

> Meridian Freight's write-heavy shipment-events DynamoDB global table has replicas in three Regions. Analysis shows the third Region receives almost no reads and is not part of any recovery plan.
>
> The recovery requirement is met by the other two Regions.

What should the solutions architect recommend?

A. Switch the table to provisioned capacity in all three Regions

B. Enable DynamoDB Accelerator in the third Region

C. Remove the third Region's replica

D. Convert to multi-Region strong consistency

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Provisioned capacity

May help, but still pays replicated writes in the unused Region.

B DAX

Adds cost in a Region with no reads.

C Remove the replica

Ends a third of the replicated write and storage cost.

D Strong consistency

Changes consistency, not cost; needs three Regions anyway.

THE SIGNAL

A replica Region nobody reads and no plan needs is pure cost.

> Meridian Freight reports costs to four business units. Each business unit owns several accounts in its own OU, but the shared platform account hosts resources for all four, tagged by CostCentre.
>
> Finance wants every report to show costs by business unit, and the mapping must survive future reorganisations.

What should the solutions architect implement?

A. Tag every resource in every account with a BusinessUnit tag

B. Move all resources into one account per business unit, including platform

C. Export the bill to spreadsheets each month

D. A cost category with rules mapping accounts to business units, and the platform account's CostCentre tags to business units

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A **Tag everything**
Huge effort, and must be redone at every reorganisation.

B **Split the platform**
Duplicates shared infrastructure for an accounting view.

C **Spreadsheets**
Manual and error-prone.

D **Cost category**
Accounts and tags mapped by rule; change rules, not resources.

THE SIGNAL

Business terms over accounts and tags, resilient to reorganisation: cost categories.

What to carry out of this module

- **Accounts first, tags second**; few mandatory keys with fixed values.
- **Activate** cost allocation tags; **backfill** up to twelve months.
- **Tag policy** for values, **SCP** for presence, **Config** for existing gaps.
- **Cost categories** turn accounts and tags into business units.
- **Replicate only what recovery needs**, in the fewest Regions, cheapest class.

> ON THE EXAM
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> Allocate structurally, enforce tags, and size replication to the recovery objective.